

Dissecting protein identity, mutation effects, and disease mechanism in ESM-2 embeddings

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Abstract

Protein language models are increasingly used to predict variant effects, including pathogenicity. Whether frozen representations can distinguish disease mechanisms under family holdout remains unevenly studied. Here we evaluate frozen ESM-2 wildtype, mutant, and delta (mutant minus wildtype) representations for classifying missense variants as loss-of-function, gain-of-function, or dominant-negative, using both gene-held-out and Pfam-family-held-out evaluation. We find that mechanism classification from the wildtype and mutant embeddings drops from gene holdout to family holdout. Wildtype embeddings also predict Pfam family well above the majority-class baseline, consistent with family recognition partly contributing to the gene-to-family drop. Under a preregistered linear probe, the delta matches the majority-class reference for this classification, while exploratory nonlinear probes recover weak performance above that reference. The same delta representation nevertheless predicts pathogenicity and folding stability under family holdout, though stability performance is family-dependent and uneven across protein domains. The pathogenicity signal is largely redundant with ESM-2’s own conservation score. Wildtype embeddings also support enzyme-type classification, confirming that the representation carries usable signal for other biological tasks. These results show that gene-held-out performance can overstate transfer to unseen protein families. Variant-effect studies should therefore report the held-out unit and use family- or homology-disjoint evaluation when claiming transfer beyond represented families.

Introduction

Protein language models such as ESM-2 are trained on large sequence databases and produce embeddings used to predict variant effects. Whether those embeddings can predict a particular downstream property is a separate empirical question (Lin et al., 2023). Performance on pathogenicity, for example, does not establish that they can distinguish disease mechanisms.

Pathogenicity and disease mechanism are distinct properties. A pathogenic variant may abolish protein function (loss-of-function), create or increase activity (gain-of-function), or interfere with the product of the wildtype allele (dominant-negative). These mechanisms can have different clinical and therapeutic consequences, so mechanism classification is separate from identifying a variant as damaging. In this study, mechanism labels are assigned at gene level from curated literature rather than direct variant-level assays. Each variant therefore inherits its gene’s label, and the analysis cannot represent mechanism differences among variants in the same gene.

Variant-level mechanism prediction could help clinical geneticists and functional genomics researchers prioritize follow-up assays and distinguish variants that may inform different therapeutic strategies. This is particularly relevant when variants in the same gene act through different mechanisms.

The unit held out during evaluation determines the generalization claim. A variant-random split tests interpolation within genes already represented during training. A gene-disjoint split tests transfer to unseen genes, but members of the same protein family can remain in training, so performance may partly reflect family-level similarity. A family- or homology-disjoint split tests transfer beyond the families represented during training.

Recent disease-mechanism predictors handle sequence relatedness differently. LoGoFunc uses a gene-disjoint primary test and a sensitivity analysis limiting training-to-test sequence identity to 40% (Stein et al., 2023). ClearVariant uses variant-level cross-validation without gene or family holdout (Ha et al., 2025), whereas PreMode evaluates variants within each target gene and also tests transfer between genes that share a domain (Zhong et al., 2025). These designs address different generalization questions, so their performance estimates should be interpreted according to the unit held out.

This study tests whether the frozen ESM-2 mutation delta supports disease-mechanism classification across held-out protein families. Before the confirmatory run reported here, we preregistered the confirmatory claims and stability-

control criteria. The record also specified the primary probes, resampling procedures, thresholds, and decision rules. Analyses outside those prospective specifications are labeled exploratory; rules added after results were inspected are identified as post-result amendments. The mechanism result is interpreted alongside three positive controls: pathogenicity, folding stability, and enzyme type. These tasks use task-specific cohorts, representations, and probes within a shared evaluation framework based on frozen ESM-2 embeddings, family-held-out testing, and cluster-aware uncertainty estimates. They provide context for the result of the preregistered mechanism probe at the majority-class reference. The study design and shared evaluation framework are summarized in Figure 1.

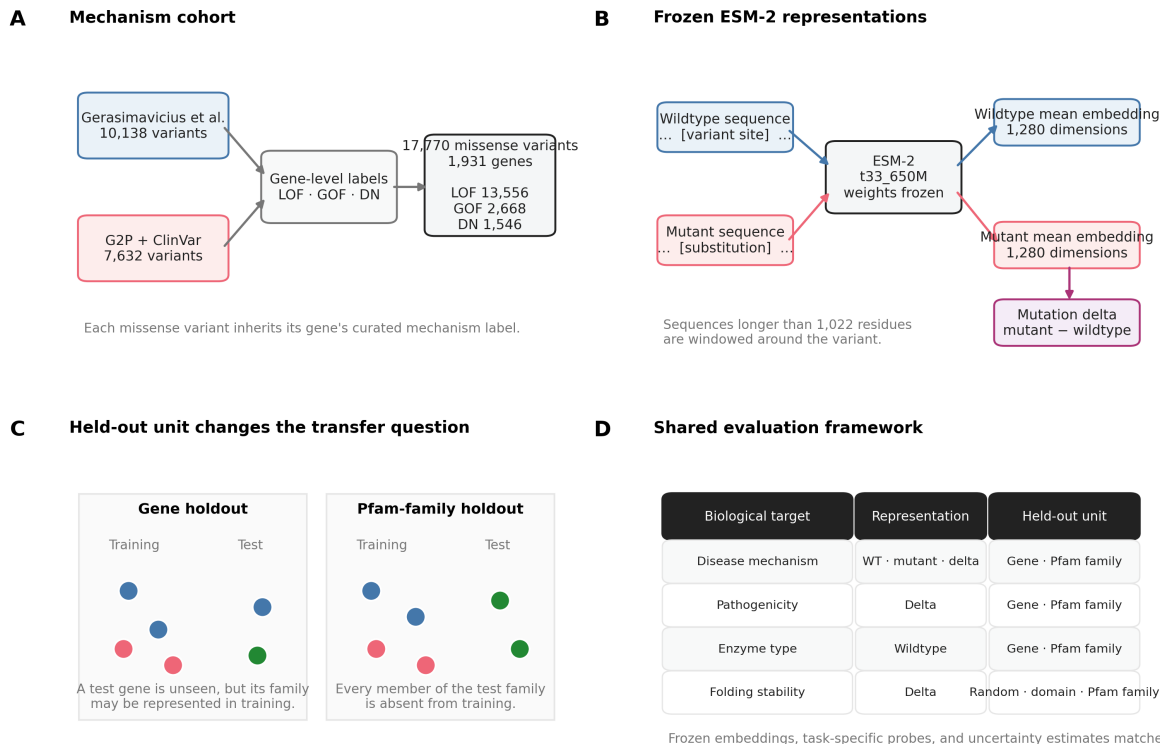


Figure 1. Study design and shared evaluation framework. (A) The mechanism cohort combined curated gene-level labels with missense variants, and each variant inherited its gene’s loss-of-function, gain-of-function, or dominant-negative label. (B) Wildtype and mutant sequences were embedded separately with frozen ESM-2. Mean-pooled wildtype, mutant, and mutation-delta representations were retained. (C) Gene holdout excluded each test gene while allowing related family members in training; Pfam-family holdout excluded all members of each test family. (D) Mechanism, pathogenicity, enzyme-type, and folding-stability analyses used task-specific representations and held-out units within the shared framework.

Results

1. The preregistered delta probe matches the majority-class reference

Under family holdout, the preregistered linear probe on the mean-pooled mutation delta scored a five-seed mean macro-F1 of 0.290, equal to the measured majority-class reference of 0.290. Across all five seeds, the upper 95% family-bootstrap confidence bound remained below 0.340, the measured reference plus the preregistered margin of 0.05. In seed 0, macro-F1 was 0.290 with a 95% confidence interval of 0.276 to 0.305. Wildtype-only and mutant-only embeddings scored above the reference but dropped when moving from gene split to family split (wildtype-only: 0.552 to 0.450; mutant-only: 0.549 to 0.451). Mutant-only performance was nearly identical to wildtype-only performance, indicating little classification benefit from using the mutant rather than the wildtype representation under this probe (Figure 2A).

A single-source analysis restricted to 10,138 Gerasimavicius variants reproduced the main pattern without the G2P additions. Using references recomputed for this subset, the delta matched the majority-class reference at reported precision under both gene holdout (0.279 versus 0.279) and family holdout (0.280 versus 0.280). Wildtype macro-

F1 decreased from 0.611 to 0.462. The main pattern therefore did not depend on adding the G2P-derived cohort (Supplementary Figure S1).

2. Weak classification and ranking signals depend on probe choice

The delta's macro one-vs-rest AUROC ranged from 0.532 to 0.578 across seeds. The five-seed family-split AUROC was 0.584 for gain-of-function, 0.557 for loss-of-function, and 0.524 for dominant-negative. Ranking performance was strongest for gain-of-function, while dominant-negative remained close to the 0.500 no-signal value. Family-block permutation tests detected ranking signal in four of five seeds ($p = 0.029, 0.003, 0.011, 0.003, 0.054$). By the preregistered three-of-five decision rule, this overturned the claim that the delta had no detectable family-robust ranking signal. Thus, although the preregistered linear probe matched the majority-class reference in macro-F1, it retained weak ranking information.

Exploratory probes recovered additional classification performance. In one analysis, a full-dimensional, standardized, class-balanced logistic regression decreased from macro-F1 0.415 under gene holdout to 0.387 under family holdout. In a separate nonlinear analysis, an MLP decreased from 0.395 to 0.375. Both family-split scores exceeded the majority-class reference of 0.290 but remained below the 0.450 obtained with the preregistered linear probe on the wildtype-only representation. These exploratory results do not replace the preregistered result. The probe comparison and class-specific ranking results are shown in Figure 2B and Figure 2C.

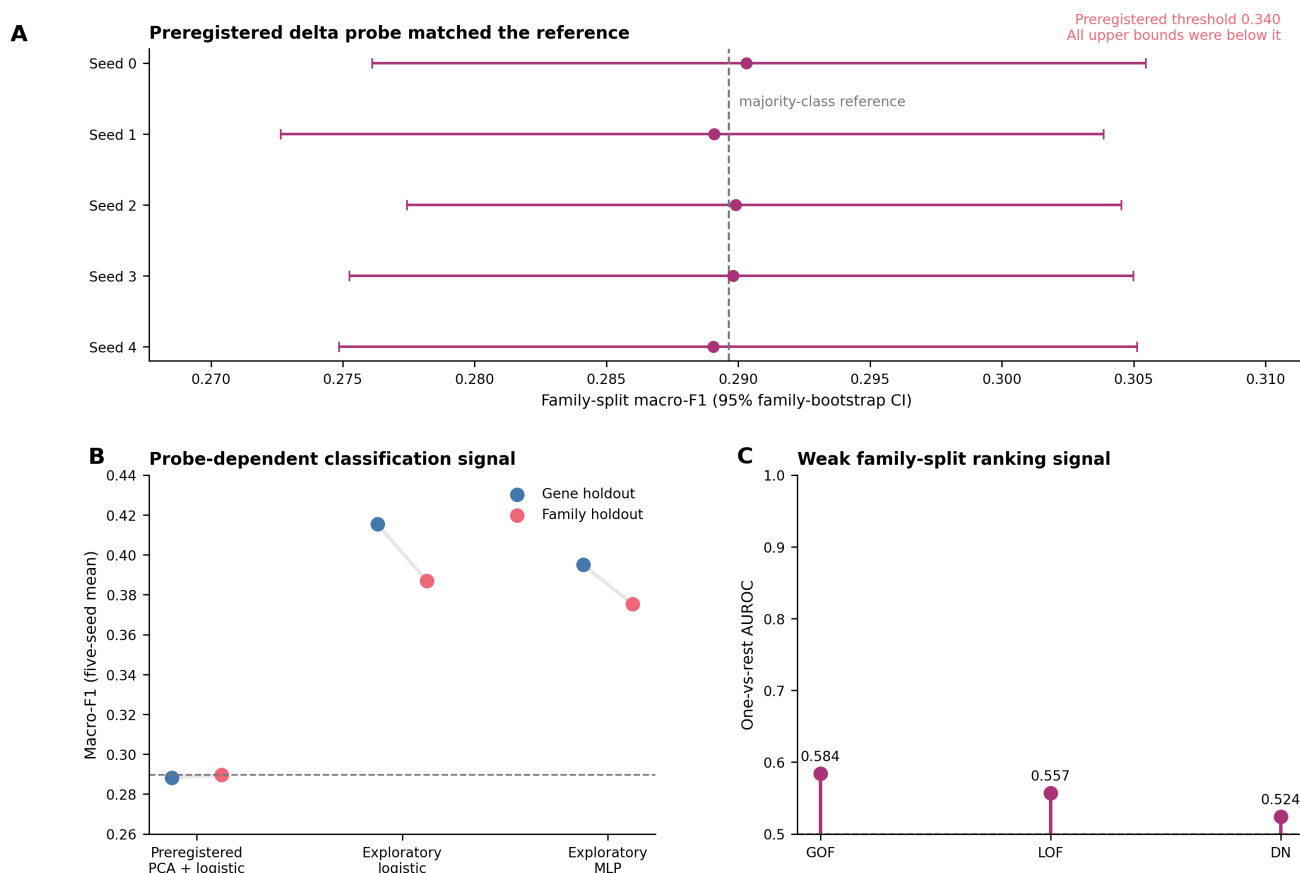


Figure 2. Disease-mechanism information in the mutation delta. (A) Family-split macro-F1 from the preregistered within-fold PCA and logistic-regression probe across five seeds. Points show out-of-fold estimates and bars show 95% family-bootstrap confidence intervals. The dashed line is the majority-class reference; the preregistered threshold of 0.340 lies beyond the displayed axis, and every upper confidence bound remained below it. (B) Five-seed mean macro-F1 under gene and Pfam-family holdout for the preregistered probe and two exploratory probe specifications. (C) Five-seed family-split one-versus-rest AUROC for gain-of-function, loss-of-function, and dominant-negative variants. The dashed line marks the 0.500 no-signal value.

3. Wildtype embeddings predict Pfam family

In a subset of 755 genes spanning 145 Pfam families, a linear probe predicted family from the wildtype embedding with a five-seed mean accuracy of 60.2%, compared with a 4.37% majority-class baseline. Applied to the mutation delta, the same probe matched the baseline at 4.37%. In a classifier-independent nearest-neighbour analysis, 25.4% of each wildtype embedding’s five nearest neighbours shared its Pfam family, compared with 0.52% after family labels were shuffled. The corresponding delta values were 5.2% and 0.52%, showing weaker but detectable family structure in the delta’s local geometry. The paired wildtype gene-to-family gap was positive in all five seeds, ranging from 0.045 to 0.139, and the corresponding 95% family-bootstrap intervals excluded zero in four of five seeds. This gap represented 38.9% of the wildtype embedding’s gene-split performance above the majority-class reference, with a 95% confidence interval of 23.9% to 54.3%. These family-readability, split-gap, and leakage-fraction analyses are shown in Figure 3.

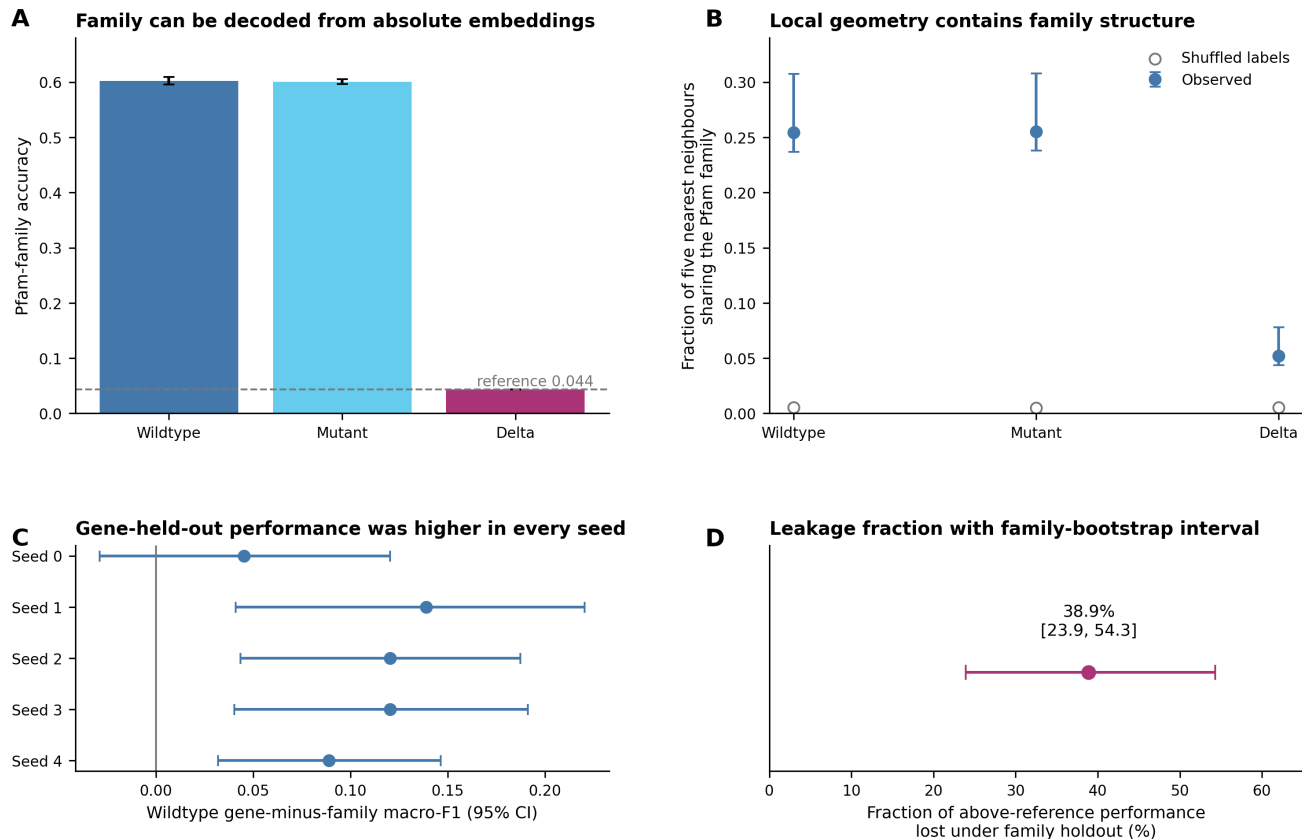


Figure 3. Pfam-family information and the gene-to-family performance gap. (A) Accuracy of a linear probe trained to classify 145 Pfam families from wildtype, mutant, or delta representations. Bars show five-seed means with standard deviations; the dashed line is the majority-class reference. (B) Fraction of each representation’s five nearest neighbours that shared its Pfam family. Filled points show observed values with 95% bootstrap confidence intervals; open points show shuffled family-label references. (C) Paired wildtype gene-minus-family macro-F1 for each seed, with 95% family-bootstrap confidence intervals. (D) Fraction of the wildtype representation’s performance above the majority-class reference that was lost under family holdout, with its 95% family-bootstrap confidence interval.

4. Pathogenicity is recoverable across families

Under family holdout, the mean-pooled mutation delta reached a five-seed mean AUROC of 0.885 with an MLP. The seed-0 estimate was 0.886 (95% CI 0.880-0.891), with the interval entirely above the preregistered threshold of 0.85. The five-seed mean decreased by only 0.003 from the gene-split result. The directional component alone yielded AUROC 0.855 with logistic regression and 0.893 with an MLP, matching or exceeding the corresponding full-delta values, while magnitude alone remained weaker at 0.610 under both probes (Figure 4A-B).

5. Pathogenicity information is largely redundant with conservation

In a matched seed-0 logistic-regression comparison, ESM-2’s masked-marginal conservation score reached AUROC 0.888, exceeding both the delta (0.835) and the delta-plus-conservation combination (0.883). Adding the delta changed AUROC by -0.005 (95% CI -0.008 to -0.001). The pathogenicity information available from the delta was therefore largely redundant with conservation in this evaluation (Figure 4C).

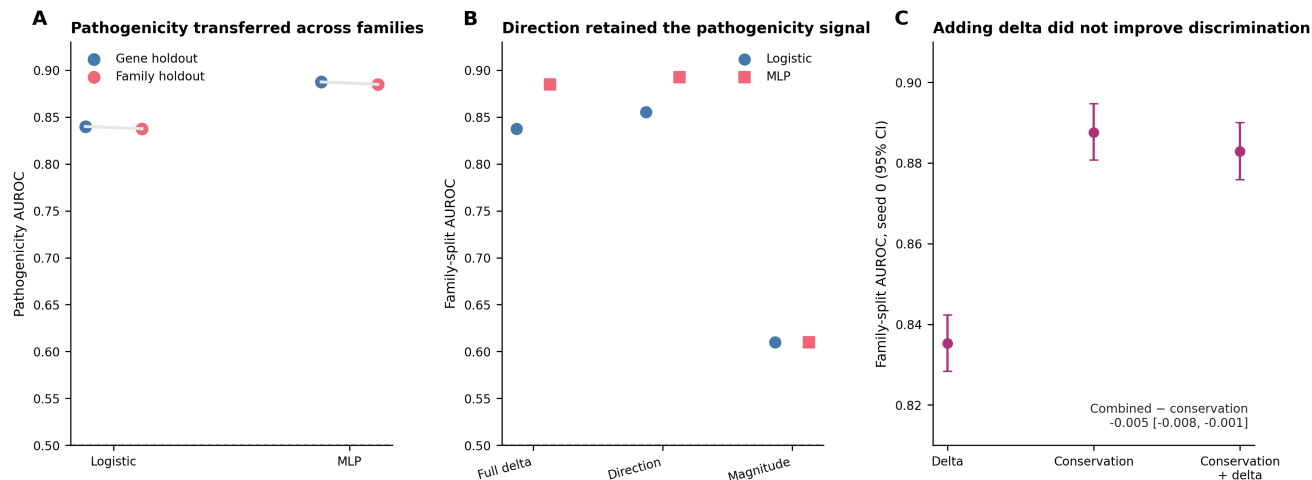


Figure 4. Pathogenicity transfer, delta geometry, and conservation. (A) Five-seed mean pathogenicity AUROC under gene and Pfam-family holdout for logistic-regression and MLP probes on the mean-pooled mutation delta. (B) Five-seed family-split AUROC from the full delta, its directional component, and its magnitude under exploratory logistic-regression and MLP probes. The dashed lines in A and B mark the 0.500 no-signal value. (C) Matched seed-0 family-split logistic-regression comparison of the delta, ESM-2 masked-marginal conservation score, and their combination. Bars show 95% family-bootstrap confidence intervals. The annotation reports the paired AUROC difference between the combined and conservation-only models.

6. Enzyme type is recoverable from the wildtype embedding

The wildtype embedding classified enzyme type as kinase, protease, oxidoreductase, or non-enzyme with a five-seed mean family-split macro-F1 of 0.779. The seed-0 estimate was 0.787 (95% CI 0.732-0.818), with the interval entirely above the preregistered threshold of 0.70 and a majority-class reference of 0.219. A proteome-feature negative control reached 0.291 under the same family holdout. On the shared set of families, the enzyme score exceeded mechanism classification by 0.507 macro-F1 (95% CI 0.447-0.541; Figure 5A,C).

Five-seed family-split AUROC was 0.957 for kinase, 0.952 for oxidoreductase, 0.942 for non-enzyme, and 0.905 for protease. Protease was the smallest class and had the least precise estimate (Figure 5B).

The paired MLP-minus-logistic difference was -0.074 (95% CI -0.118 to -0.043). The interval was not wholly contained within the preregistered equivalence range of -0.05 to +0.05, so linear-nonlinear equivalence was not established (Figure 5C).

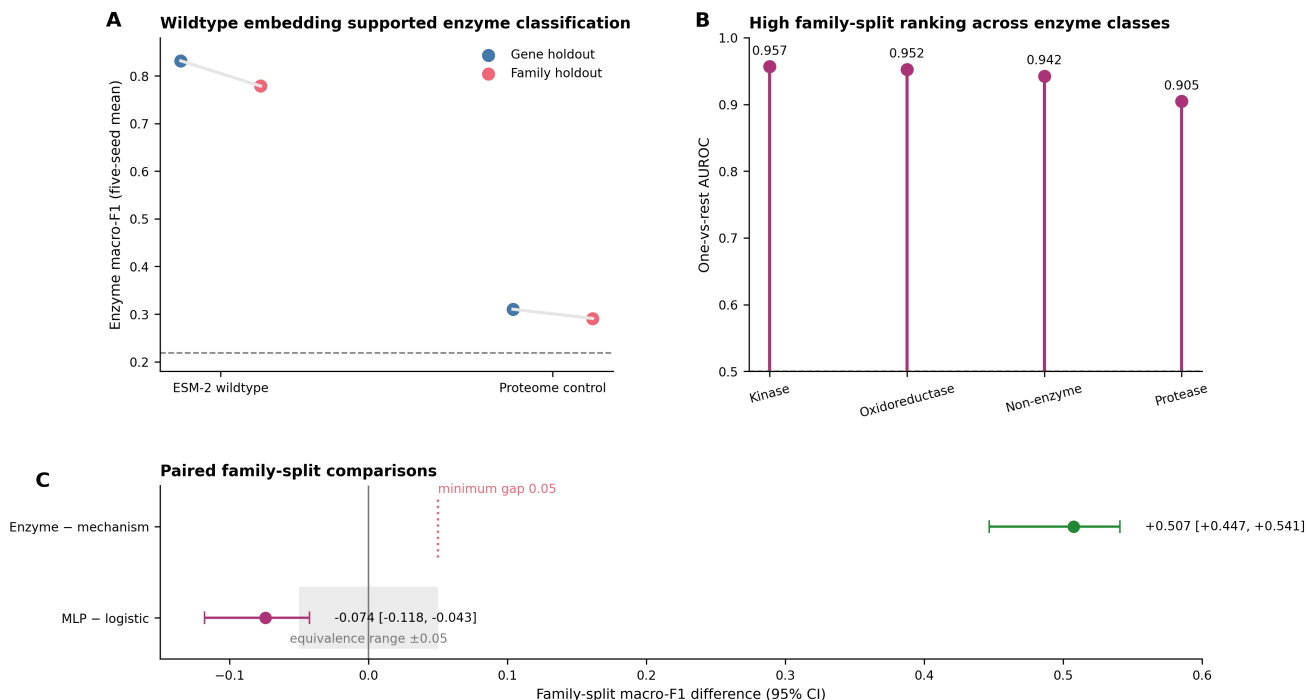


Figure 5. Enzyme-type classification under family holdout. (A) Five-seed mean macro-F1 under gene and Pfam-family holdout for the ESM-2 wildtype embedding and the proteome-feature negative control. The dashed line is the majority-class reference. (B) Five-seed family-split one-versus-rest AUROC from the logistic-regression probe for each enzyme class. The dashed line marks the 0.500 no-signal value. (C) Paired family-split comparisons. The upper row shows enzyme-minus-mechanism macro-F1 on the shared family subset, with the comparison minimum gap marked by the dotted line. The lower row shows MLP-minus-logistic enzyme macro-F1, with the equivalence range shaded. Points show paired differences and bars show 95% family-bootstrap confidence intervals.

7. Stability is recoverable but not uniformly family-robust

The mean-pooled mutation delta predicted experimentally measured folding stability. Under a random split, the seed-0 Spearman correlation from the preregistered ridge probe was 0.693 (95% CI 0.675-0.709), with the interval entirely above the threshold of 0.50. Under family holdout, the five-seed mean fell to 0.554. On the matched seed-0 cohort, the random-to-family decrease was 0.153 (95% CI 0.112-0.192), with the interval entirely above the preregistered 0.10 boundary for family dependence. Performance also varied among domains: the per-domain correlation standard deviation was 0.160 (95% CI 0.132-0.183), above the registered maximum of 0.10 (Figure 6A-C).

Using the exploratory full-dimensional, standardized, class-balanced mechanism classifier, removing a fitted stability direction changed mechanism macro-F1 by -0.0009 (95% CI -0.0025 to +0.0007; Supplementary Figure S2). Exploratory nonlinear probes retained family-held-out stability signal, with correlations of 0.627 for the MLP and 0.630 for XGBoost, but did not alter the preregistered linear-probe conclusions.

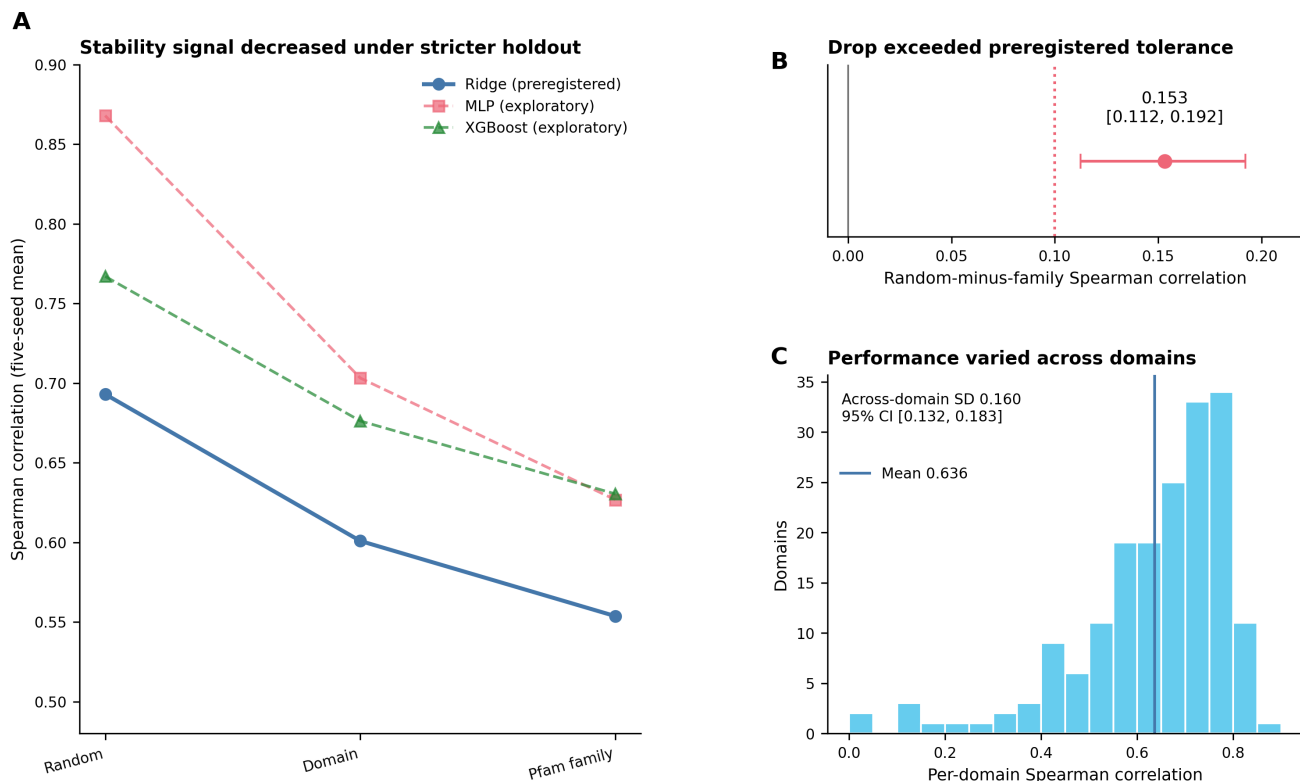


Figure 6. Folding-stability transfer across held-out units. (A) Five-seed mean Spearman correlation under random, PDB-domain, and Pfam-family holdout. Ridge regression was the preregistered probe; MLP and XGBoost were exploratory. (B) Matched seed-0 random-minus-family difference for the ridge probe, with its 95% family-bootstrap confidence interval. The dotted line marks the preregistered 0.10 tolerance for family dependence. (C) Distribution of per-domain Spearman correlations. The vertical line marks the across-domain mean, and the annotation reports the standard deviation and its 95% bootstrap confidence interval.

Methods

1. Mechanism cohort and labels

Gene-level disease-mechanism labels were assembled from Gerasimavicius et al. (2022) and Gene2Phenotype (G2P; Thormann et al., 2019). Gerasimavicius assignments took priority. G2P records were restricted to strong or definitive gene-disease associations with an unambiguous molecular-mechanism annotation; unresolved conflicts were excluded. Haploinsufficiency and autosomal-recessive annotations were combined as loss-of-function (LOF); the other classes were gain-of-function (GOF) and dominant-negative (DN). Labels were gene-level, so each variant inherited its gene’s label and the LOF class was not equivalent to a variant-level functional-assay classification.

Missense variants came from the Gerasimavicius supplementary data and from pathogenic ClinVar records (Landrum et al., 2018) for genes covered only by G2P; likely-pathogenic records were excluded. Variants required a UniProt identifier (The UniProt Consortium, 2023), an available reference sequence, a complete substitution, and agreement between the recorded and reference wildtype residues. The final cohort contained 17,770 variants across 1,931 genes: 13,556 LOF, 2,668 GOF, and 1,546 DN. Pfam annotations covered 1,907 genes in 1,144 families.

2. Pathogenicity cohort

The pathogenicity control used a separate ClinVar extraction from the same target-gene universe (Landrum et al., 2018). The extraction retained GRCh38 single-nucleotide missense records. Records classified as conflicting, uncertain, not provided, or other, and records without assertion criteria, were excluded. Pathogenic and likely-pathogenic records formed the pathogenic class; benign and likely-benign records formed the benign class.

Protein-level substitutions were deduplicated, and one substitution with conflicting class labels was excluded. Within each gene, up to 20 variants per class were sampled with seed 42 and reduced to equal class counts. Variants that could not be applied to the UniProt reference were removed, after which per-gene balance was restored. The final cohort contained 24,384 variants across 1,802 genes, with 12,192 variants per class. Family splits used 24,176 variants in 1,072 Pfam clusters; 208 variants without Pfam assignments were excluded only from those splits.

3. Enzyme cohort

Enzyme-type labels were derived from UniProt EC-number and keyword annotations (The UniProt Consortium, 2023). Genes were classified as kinase, protease, oxidoreductase, or non-enzyme, with priority in that order when annotations overlapped. Enzymes outside the three named classes and genes without the required annotation were excluded. The cohort contained 1,451 genes: 130 kinases, 68 proteases, 123 oxidoreductases, and 1,130 non-enzymes. Here, non-enzyme meant no annotated EC activity. The primary representation was the wildtype embedding; a 33-feature proteome matrix was the prespecified negative control. Family splits included 1,429 genes in 835 Pfam clusters for the embedding and 1,422 genes in 828 clusters for the aligned proteome features.

4. Stability cohort

Folding-stability measurements were obtained from the Tsuboyama et al. (2023) mega-scale assay. Single substitutions in natural PDB domains with numeric `ddG_ML` values were retained; designed proteins, mutated backgrounds, insertions, deletions, and multi-mutants were excluded. The target was the reported folding-stability change, $\Delta\Delta G$, for 177,315 variants across 181 domains. Evaluation used random, domain-disjoint, and Pfam-family-disjoint splits. HMMER searches (Eddy, 2011) against Pfam-A (Mistry et al., 2021) assigned the best hit passing the curated gathering threshold. Fourteen unassigned domains were excluded only from the family split, leaving 167 domains in 77 families.

5. ESM-2 representations

Wildtype and mutant sequences were embedded separately with frozen `esm2_t33_650M_UR50D`. Final-layer residue representations were mean-pooled into 1,280-dimensional vectors, and the mutation delta was mutant minus wildtype. The delta at the substituted residue was also retained. Sequences longer than 1,022 residues were windowed around the variant. The pathogenicity conservation score was $\log P(\text{wildtype}) - \log P(\text{mutant})$ at the masked variant position.

6. Evaluation splits

Gene splits kept all variants from a gene in one fold; family splits kept all genes with the same assigned Pfam accession together. Mechanism, pathogenicity, and enzyme analyses used the first Pfam cross-reference returned by UniProt and excluded unassigned genes only from family splits. Stability used the HMMER assignments above and also held out complete PDB domains. Five-fold cross-validation was repeated over five seeds. Metrics were computed within folds and averaged. Confidence intervals used 1,000 cluster-bootstrap resamples: genes for gene splits, Pfam families for family splits, and PDB domains for stability random and domain splits.

7. Probes

The preregistered mechanism probe fitted up to 256 principal components and logistic regression within each training fold, without standardization or class weighting. Exploratory mechanism probes included full-dimensional, fold-standardized, class-balanced logistic regression and a two-hidden-layer multilayer perceptron (MLP). Pathogenicity used fold-standardized logistic regression and a one-hidden-layer MLP; enzyme classification used logistic regression and the two-layer MLP. The preregistered stability probe was fold-standardized ridge regression with an L2 penalty of 1.0. Other stability models were exploratory. No classifier estimated calibrated clinical risk. Full hyperparameters are reported in Supplementary Methods.

8. Preregistration and statistics

The statistical plan was written and committed before the confirmatory run and is retained with its revision history in the version-controlled analysis repository. It specified the prospective claims, stability controls, primary probes, thresholds, resampling procedures, and decision rules. Analyses outside those specifications were exploratory. Dated post-result amendments recorded protein-level deduplication and seed-0 inference for pathogenicity, seed-0 inference for the conservation comparison, and the numerical enzyme-comparison threshold. These rules were not prospective; the affected analyses were rerun under the amended specifications.

Primary metrics were macro-F1 for multiclass classification, one-versus-rest AUROC for class ranking, binary AUROC for pathogenicity, and Spearman correlation for stability. Always-LOF and always-non-enzyme classifiers supplied the multiclass references; AUROC and Spearman no-signal values were 0.500 and 0.000. Mechanism ranking used 1,000 family-block label permutations per seed. The mechanism classification, ranking, and gene-to-family claims used their prespecified three-of-five seed rules. Other threshold tests used seed-0 out-of-fold predictions and cluster-bootstrap intervals for inference; five-seed means were descriptive.

9. Use of generative AI

Generative AI tools, including Claude (Anthropic) and Codex (OpenAI), were used to assist with code development, analysis review, figure preparation, and manuscript drafting and editing. All outputs were reviewed and verified by the author, who takes full responsibility for the work.

Discussion

A central finding was that mechanism-classification performance from frozen ESM-2 wildtype and mutant embeddings dropped when evaluation moved from held-out genes to held-out Pfam families. Wildtype embeddings also predicted Pfam family well above the majority-class baseline, despite containing no mutation information. Together, these observations suggested that gene-held-out performance may partly reflect family-specific associations learned from other members of the same family, which were absent under family holdout. The wildtype mechanism classifier performed better under gene holdout in all five seeds, and the family-bootstrap intervals excluded zero in four of five seeds. However, these analyses did not show that family membership predicted mechanism labels beyond class imbalance, so they did not establish family recognition as the cause of the gap.

Under family holdout, the preregistered linear probe on the mutation delta matched the majority-class reference for three-class disease-mechanism classification. This did not establish that the delta contained no mechanism information. Permutation testing detected ranking signal in four of five seeds, and exploratory probes recovered modest above-reference classification. These findings together indicated weak, probe-dependent signal but did not change the preregistered classification result.

The mutation delta predicted pathogenicity and folding stability, while the wildtype embedding predicted enzyme type. These controls showed that the shared evaluation framework could recover biological signals from frozen ESM-2 embeddings under family holdout. On the shared family subset, enzyme classification exceeded mechanism classification by 0.507 macro-F1, with a paired 95% bootstrap interval from 0.447 to 0.541. Because the controls used task-specific cohorts, representations, and probes, they did not establish equal task difficulty or label quality.

The pathogenicity signal from the mutation delta was substantially redundant with ESM-2’s masked-marginal conservation score. Conservation alone outperformed the delta, and adding the delta slightly reduced discrimination. The delta’s direction retained nearly all pathogenicity performance, whereas its magnitude was much weaker. This comparison did not establish that conservation is the delta’s only content. Zhong and Shen (2022) report a related zero-shot contrast: ESM1b reaches AUROC 0.922 for pathogenicity but 0.541 to 0.653 for GOF-versus-LOF classification across four families. Their trained RESCUE model improves on the zero-shot score in three of the four tasks, and they argue that mechanism prediction should be family-specific. Their evaluation therefore tests prediction within known families rather than transfer to unseen families.

Folding stability remained predictable under family holdout, but both the random-to-family decline and cross-domain variability exceeded their preregistered limits. Exploratory nonlinear probes improved absolute performance but did not change these linear-probe conclusions. Mutation-level information therefore remained usable while transferring unevenly across protein families.

Existing mechanism predictors use different held-out units. Among the studies reviewed, LoGoFunc is the closest precedent, with a gene-disjoint primary test and a sensitivity analysis limiting training-to-test sequence identity to 40% (Stein et al., 2023). ClearVariant uses variant-level cross-validation without gene or family holdout (Ha et al., 2025), whereas PreMode primarily evaluates variants within each target gene and also tests transfer between genes sharing a domain (Zhong et al., 2025). These designs address different generalization questions. The present study directly compares gene and family holdout, measures family information in the representation, and uses positive controls within the shared evaluation framework. The literature review was targeted rather than systematic and does not establish priority for this combination.

The main limitation is label granularity. Mechanism labels are curated per gene and applied to individual variants, so the analysis cannot represent within-gene heterogeneity and the assigned label may not describe the mechanism of every variant. Badonyi and Marsh (2025) report that 43% of multi-phenotype dominant genes and 49% of mixed-inheritance genes carry both loss-of-function and non-loss-of-function mechanisms. The dominant-negative class is also the smallest, reducing the precision of class-specific estimates.

Four further constraints apply. One Pfam accession per protein does not eliminate remote homology. The representation analyses use one frozen ESM-2 model size and emphasize mean pooling. The pathogenicity control shares the mechanism cohort’s target-gene universe, making it a matched comparison rather than an independent replication. Protein-level deduplication and seed-0 inference for the pathogenicity and conservation analyses were specified after initial results were inspected, and those analyses were rerun under documented post-result amendments.

Evaluations of mechanism prediction in unseen proteins should report the held-out unit and an appropriate no-signal reference alongside performance. The next priority is variant-level mechanism labels established by functional assays. Such labels would address label granularity; stricter homology partitions would test transfer; and additional model sizes and pooling strategies would test representation choice.

Author contributions

Dami Gupta conceived the study, developed the methodology and software, curated the data, performed the analyses, prepared the figures, and wrote and revised the manuscript.

Code availability

Code used for cohort construction, analysis, and figure generation is available at https://github.com/dami-gupta-git/esm2_mechanism.

Data availability

The reproducibility package supporting this study is available in Zenodo at <https://doi.org/10.5281/zenodo.22037471>. It contains the processed cohorts, statistical outputs, out-of-fold predictions, reports, figures, preregistration records, environment records, and frozen source snapshot. Large ESM-2 embedding arrays and upstream download caches are excluded; row-identity metadata, sequence inputs, model identifiers, and content fingerprints are included.

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Competing interests

The author declares no competing interests.

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